

Dynamics of Newtonian and relativistic thin fluid shells

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Abstract

We study the state space structure of both the Newtonian and relativistic equations of motion describing thin fluid shells, and identify and classify the fixed points of their dynamics. Using concepts borrowed from dynamical systems theory, we provide an alternative proof of a general condition for the existence of Newtonian shell oscillations. For the case of polytropic relativistic shells, we numerically determine the stability region of the parameter space that supports shell oscillations.

Keywords: thin shells, dynamical systems, Newtonian gravity, general relativity

1 Introduction

Thin shells are idealized matter distributions restricted to a spatial surface whose dynamics are governed by the competing forces of their self-gravity and surface pressure. There is a long tradition within gravitational physics of studying the equilibrium properties and dynamical behavior of these systems to better understand gravitational phenomena whose underlying physics might be obscured in more complicated and realistic matter models [1, 2]. As an example, thin dust (i.e. pressureless) shell collapse is now a textbook example in general relativity, and is one of the simplest settings in which to study the collapse of material body into a black hole [3]. Recently, LeMaitre and Poisson [4] have exploited the simplicity of thin fluid shells to explain the onset of dynamical instability when stars reach their maximum mass along a sequence of equilibrium solutions.

In this paper, we study the Newtonian and relativistic thin shell equations from a dynamical systems viewpoint, which is known to provide qualitative but powerful insights into the global behavior of the solutions of differential equations. With these methods, we are able to shed light on the dynamics of these shells for all possible initial conditions. We also prove a few statements concerning the stability of shell equilibria, and in particular, we numerically compute the stability region for polytropic relativistic shells.

2 Newtonian shell dynamics

We first consider the equations of motion of Newtonian thin shells. There are two forces to account for here: the outward force due to surface pressure, and the inward gravitational force that the shell exerts on itself. The surface pressure p is related to the work W done by the shell as it changes its area A by the equation $dW = p dA$. On the other hand, we have $dW = F dR$, so we obtain a total force due to surface pressure of $8\pi pR$. The total gravitational force has a magnitude of $Gm^2/2R^2$. The resulting equation of motion can then be recast in first-order form as

$$\dot{R} = V, \quad \dot{V} = -\frac{Gm}{2R^2} + \frac{8\pi p(\sigma(R))R}{m}, \quad (1)$$

where $R(t)$ is the radius of the shell, $V(t)$ is its velocity, m is the total mass of the shell and is assumed to be constant, $\sigma := m/4\pi R^2$ is the density of the shell, and $p = p(\sigma(R))$ is the equation of state. This is an autonomous system of differential equations, and provides a time-independent vector field on the two-dimensional (R, V) state space.

We begin our analysis by transforming to nondimensionalized variables. We let $R = r_c r$, $t = t_c \tau$, and $V = v_c u$, choosing $r_c = R_0 =: R(0)$, $t_c = \sqrt{2R_0^3/Gm}$, $v_c = \sqrt{Gm/2R_0}$ as characteristic length, time, and velocity scales of the system. Equation (1) then becomes

$$\frac{dr}{d\tau} = u, \quad \frac{du}{d\tau} = -\frac{1}{r^2} + \varepsilon(r)r, \quad (2)$$

where $\varepsilon(r) := 16\pi p(\sigma(r))R_0^3/Gm^2$ carries the equation of state.

We define $A := 16\pi R_0^3/Gm^2$, and we assume that $P = (r_*, u_*)$ is a fixed point of this system. It follows that $u_* = 0$ and $p(\sigma(r_*)) = 1/Ar_*^3$. Close to P , Eq. (2) can be linearized into $\dot{\mathbf{v}} = J\mathbf{v}$, with $\mathbf{v} := (r - r_*, u - u_*)^T$, and J the Jacobian matrix evaluated at P . Explicitly, the Jacobian is

$$J = \begin{pmatrix} 0 & 1 \\ \frac{2}{r_*^3} + Ap(\sigma(r_*)) + Ap'(\sigma(r_*))r_* & 0 \end{pmatrix}, \quad (3)$$

where the prime $'$ denotes differentiation with respect to r . For the system to have a fixed point that is a center at P , we require that

$$\frac{2}{r_*^3} + Ap(\sigma(r_*)) + Ap'(\sigma(r_*))r_* < 0, \quad (4)$$

as this condition implies that $\det(J) > 0$. Using the chain rule, this inequality simplifies to

$$\left. \frac{dp}{d\sigma} \right|_{r=r_*} > \frac{3Gm}{8R_0 r_*}. \quad (5)$$

Moreover, we define a function $\gamma(\sigma)$ by the relation $dp/p = \gamma d\sigma/\sigma$. We will refer to $\gamma(\sigma)$ as the effective polytropic exponent for the equation of state. With this, the condition of Eq. (5) simplifies to $\gamma(\sigma(r_*)) > 3/2$. When this is satisfied, the fixed point P is a center for the system. By reversing the inequality in Eq. (4) to determine when $\det(J) < 0$, we find that the condition for P to be a saddle point is $\gamma(\sigma(r_*)) < 3/2$. In Figure 1, we plot a few phase portraits for equations of state of polytropic form, which have $p = K\sigma^\gamma$ for positive constants K and γ .

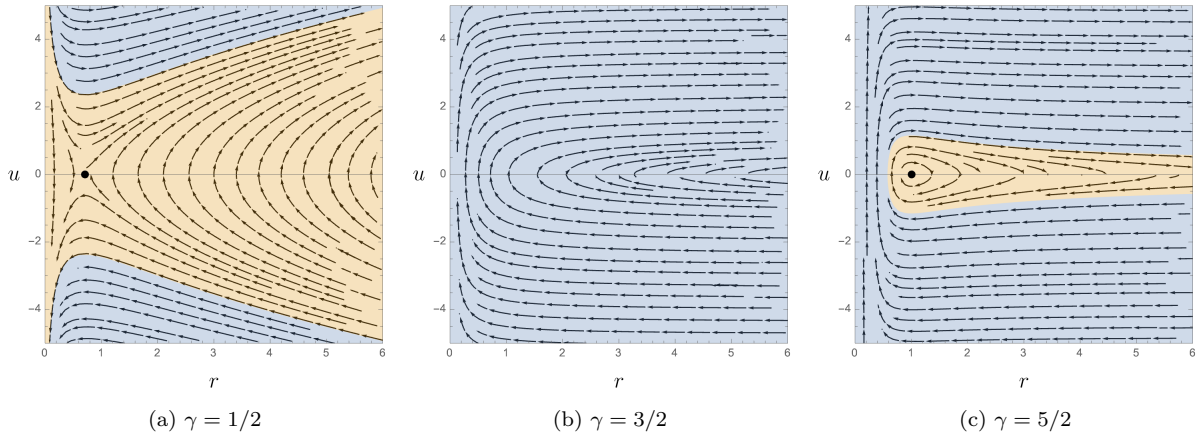


Figure 1: Phase portraits for different polytropic indices. The blue and orange regions correspond to states with $E > 0$ and $E < 0$, respectively, and the shaded circles indicate fixed points. Note that while these indices give qualitatively different behavior, all three allow for “bounce” solutions, where the shell contracts to a minimum radius before expanding outward indefinitely. Observe also that in Fig. 1a, shells with $E > 0$ either contract or expand indefinitely.

Of course, we recognize that Eq. (2) is a simple conservative system that can also be analyzed by elementary means. It is not difficult to verify that the phase paths are represented by the equation $E = u^2/2 + U_{\text{eff}}(r)$, where $U_{\text{eff}}(r) = -1/r - U(r)$ and $U(r) = \int \varepsilon(r)r dr$. For a polytropic equation of state, $\varepsilon(r) \sim r^{-2\gamma}$, so $U(r) \sim r^{2(1-\gamma)}/(2(1-\gamma))$ for $\gamma \neq 1$ and $U(r) \sim \log(r)$ for $\gamma = 1$. For $0 < \gamma < 1$, $U(r)$ is positive and vanishes as $r \rightarrow 0$. Turning points (i.e., where $u = 0$) are then only possible when $E < 0$. In Fig. 1a, the phase paths that cross the abscissa satisfy this condition.

When $\gamma > 1$, we have $U(r) < 0$, and there are two cases to consider. For $1 < \gamma < 3/2$, all $E < 0$ paths cross the abscissa only once; this is only possible for small shells. Paths for which $E > 0$ cross either twice or not at all. Above a critical positive energy, there is no turning point, and the shell either contracts to $r = 0$ or expands indefinitely. When there are two crossings, they correspond to two branches of a topologically hyperbolic curve, one branch for shells smaller than the fixed point radius, and the other for larger shells. The roles are somewhat reversed for $\gamma > 3/2$; in this case, all $E > 0$ paths cross the abscissa only once, corresponding to “scattering” dynamics for the shell. Meanwhile, any kinematically accessible path with $E < 0$ has two turning points, and represents an oscillating shell.

3 Relativistic shell dynamics

The equations of motion for a thin spherical shell in general relativity follow from conditions that need to be imposed when matching two spacetimes across a spacelike boundary, i.e. the thin shell. Inside the shell, the spacetime is Minkowski, and outside it is Schwarzschild with mass parameter M . Reference [3] can be consulted for the derivation and use of this junction formalism.

The equations of motion take the form [4]:

$$\dot{R} = V \quad (6a)$$

$$\dot{V} = -\sqrt{1+V^2} \left(\frac{m}{2R^2} - \frac{8\pi p(\sigma(R, m))R}{m} \left(\sqrt{1+V^2} - \frac{m}{R} \right) \right) \quad (6b)$$

$$\dot{m} = -8\pi p(\sigma(R, m))RV, \quad (6c)$$

where the overdot signifies a derivative with respect to proper time τ (recorded by a clock that is comoving with the shell), $R(\tau)$ is the radius of the shell, $V(\tau)$ is its velocity, m is the *material mass* of the shell, $\sigma := m/4\pi R^2$ is the density of the shell, and $p = p(\sigma(R, m))$ is the equation of state. These equations are expressed in geometric units in which $G = c = 1$. Immediate differences with the Newtonian case are the use of proper time, the more complicated “acceleration” on the right hand side of Eq. (6b), and the non-constancy of m from Eq. (6c) on account of the fact that in general relativity, the material mass includes the internal energy of the shell, which changes because of the work done by the pressure. If the shell were made of pressureless matter, this material mass would be constant. The flow of Eq. (6) preserves what is known as the gravitational mass, $M := m\sqrt{1+\dot{R}^2} - m^2/2R$, which is the mass parameter of the external Schwarzschild spacetime. Physically, this constraint says that the shell’s gravitational mass comes from its relativistic kinetic energy (the first term) and its binding energy (the second term). Mathematically, this constraint implies that the (R, V, m) state space is foliated by two-dimensional integral manifolds labeled by M . Dynamics is confined to only one of these surfaces, which in turn is selected by initial conditions.

We can nondimensionalize Eq. (6) with M , which serves as our natural length, time, and mass scale, because of our use of geometric units. We then set $R = Mr$, $\tau = M\bar{\tau}$, and $m = M\bar{m}$, and write the differential equations in the form

$$r' = v \quad (7a)$$

$$\bar{m}' = -\varepsilon(r, \bar{m})rv \quad (7b)$$

$$v' = -\sqrt{1+v^2} \left(\frac{\bar{m}}{2r^2} - \frac{\varepsilon(r, \bar{m})r}{\bar{m}} \left(\sqrt{1+v^2} - \frac{\bar{m}}{r} \right) \right), \quad (7c)$$

where the prime ' denotes differentiation with respect to $\bar{\tau}$, and $\varepsilon(r, \bar{m}) := 8\pi Mp(r, \bar{m})$. This is subject to the constraint

$$1 = \bar{m}\sqrt{1+v^2} - \frac{\bar{m}^2}{2r}, \quad (8)$$

which is the conservation of M statement expressed in terms of our dimensionless variables. Clearly, the dynamics on all constant- M surfaces differ only by a rescaling of the pressure. In what follows, we shall only consider polytropic equations of state: $\varepsilon = K\sigma^\gamma \equiv k(\bar{m}/r^2)^\gamma$.

Expressing $\bar{m}$ in terms of r and v in Eq. (8) and inserting this into Eq. (7) returns the expressions for the vector field on the reduced (r, v) state space. There are two possibilities for $\bar{m}$, which we represent as

$$\bar{m} = g_\pm(r, v) := r\sqrt{1+v^2} \pm \sqrt{r(-2+r(1+v^2))}. \quad (9)$$

The system is kinematically restricted to $r \in (2/(1+v^2), \infty)$.

The reduced vector field therefore takes two forms:

$$r' = v \quad (10a)$$

$$v' = -\sqrt{1+v^2} \left(\frac{g_\pm(r, v)}{2r^2} - \frac{\varepsilon(r, g_\pm(r, v))r}{g_\pm(r, v)} \left(\sqrt{1+v^2} - \frac{g_\pm(r, v)}{r} \right) \right) =: f_\pm(r, v), \quad (10b)$$

and it is clear that fixed points are $P = (r_*, 0)$, where r_* is a solution of $f_\pm(r_*, 0) = 0$. Now

$$f_\pm(r, 0) = \frac{1}{2r} \left(-1 \mp \sqrt{1 - \frac{2}{r}} \right) \mp k \left(r \pm \sqrt{r(r-2)} \right)^{\gamma-1} r^{1-2\gamma} \sqrt{1 - \frac{2}{r}}. \quad (11)$$

When $v = 0$, r is restricted to $(2, \infty)$. One can verify numerically that f_+ has no positive root, whereas f_- can have either zero, one, or two positive roots. Regions in the (k, γ) parameter space with no roots do not support any shell equilibria. When there is only one root, it is an unstable equilibrium, and if there are two roots, then the larger of the two roots will be stable (i.e. the larger shell). This can easily be understood by considering the behavior of $f_-(r, 0)$ and invoking a few continuity arguments. At $r = 2$, $f_-(r = 2, 0) = -1/4 < 0$. Therefore, the first root encountered as one increases r from $r = 2$, call this r_* , is a crossing of f_- from a negative to a positive value. This is an unstable fixed point, because for some small $\delta > 0$, we have $f_-(r_* - \delta, 0) < 0$, which according to Eq. (10b) “pulls” or contracts the shell away from r_* . Similarly, $f_-(r_* + \delta, 0) > 0$ means that a shell slightly larger than r_* will expand further away from equilibrium. This is equivalent to the condition $df_-/dr|_{r=r_*} > 0$. To get a stable fixed point, then, f_- needs to have a second root where $df_-/dr|_{r=r_*} < 0$. Thus, phase portraits possessing a center point will always also have a saddle point. All of these statements are consistent with conditions based on the determinant of the Jacobian matrix of the system in Eq. (10).

In Fig. 2a, which is for $(k = 2, \gamma = 1)$, we see the phase portrait surrounding a single fixed point. As anticipated, this shows an unstable saddle. A typical phase portrait with two fixed points is illustrated in Fig. 2c, which corresponds to $(k = 20, \gamma = 3)$. Note that it has a saddle (unstable shell) and a center (stable shell). In Fig. 2b, we show the stability region in the (k, γ) parameter space, which was computed using Mathematica. For a given (k, γ) , the positive real roots of Eq. (11) were solved for numerically, and the Jacobian was evaluated at each of these roots. If these parameters admitted a stable center (i.e. $\det(J) > 0$ for at least one of the roots), the corresponding point was colored blue.

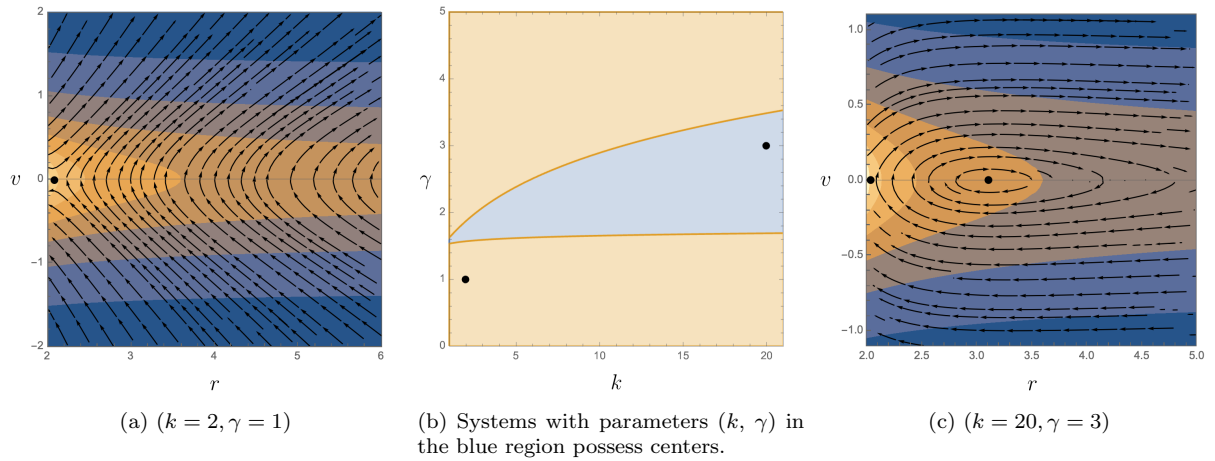


Figure 2: Relativistic shell phase portraits and numerically computed parameter region of stability. The coloring in Figs. 2a and 2c gives a rough idea of $\bar{m} = g_-(r, v)$ in the region, with warmer colors corresponding to larger $\bar{m}$. As in Fig. 1, the shaded circles indicate fixed points. While it is not evident at this scale, the fixed point closer to $r = 2$ in Fig. 2c is an unstable saddle. Finally, Fig. 2b suggests that $\gamma > 3/2$ is a necessary condition for a polytropic equation of state to support a center.

4 Conclusion and outlook

We have relied on concepts from dynamical systems theory to elucidate the state space structure of self-gravitating thin shells in Newtonian theory and general relativity. Along the way, we also derived a number of key results concerning their stability. Among these was a plot depicting the parameter region of stability for polytropic relativistic shells. Interestingly, this showed that very soft and very stiff equations of state do not support stable equilibria. Moving forward, we hope to establish some of these numerical findings on a more rigorous mathematical basis.

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